

Week 8: Review of Weeks 1–7

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October 31, 2025

Session 8.1 Homework

Instructions: Show all work. Circle your final answers.

Topics Covered:

- Week 1: Algebra Foundations
- Week 2: Solving Equations
- Week 3: Inequalities
- Week 4: Linear Equations
- Week 5: Systems of Linear Equations
- Week 6: Linear Functions & Lines
- Week 7: Systems Applications & Inequalities

Quick Reference: Key Formulas & Concepts

Order of Operations (PEMDAS)

1. Parentheses
2. Exponents
3. Multiplication/Division (left to right)
4. Addition/Subtraction (left to right)

Properties of Real Numbers

Distributive: $a(b + c) = ab + ac$

Identity: $a + 0 = a$, $a \cdot 1 = a$

Slope-Intercept Form

$y = mx + b$ where m = slope, b = y-intercept

Point-Slope Form

$$y - y_1 = m(x - x_1)$$

Distance Formula (Motion)

$$d = rt \quad (\text{distance} = \text{rate} \times \text{time})$$

Mixture Problems

$$(\text{concentration}_1)(\text{volume}_1) + (\text{concentration}_2)(\text{volume}_2) = (\text{concentration}_{\text{final}})(\text{volume}_{\text{final}})$$

Part A: Foundations (Order of Operations & Properties)

Problem 1

Simplify: $8 + 12 \div 4 - 2 \times 3$

Problem 2

Simplify: $(15 - 3)^2 \div 6 + 4$

Problem 3

Simplify by combining like terms: $8x - 3y + 2x + 7y - 5$

Problem 4

Simplify: $4(2x - 3) - 3(x + 5) + 7$

Problem 5

A rectangular garden has length $(3x + 5)$ feet and width $(2x - 1)$ feet.

Write a simplified expression for the perimeter.

Part B: Equations

Problem 6

Solve: $5x - 8 = 2x + 13$

Problem 7

Solve: $\frac{1}{3}(x - 6) = 0.5x + 4$

Problem 8

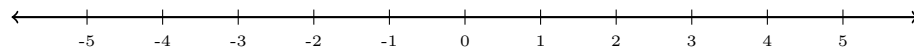
Solve for h : $A = \frac{1}{2}bh$

Part C: Inequalities

Problem 9

Solve: $-4x + 7 \leq -9$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Problem 10

Solve: $-5 < 2x + 1 \leq 7$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Problem 11

A streaming service costs \$8.99 per month.

You have a \$50 gift card.

Write and solve an inequality to find the maximum number of whole months you can subscribe.

Part D: Linear Functions & Lines

Problem 12

Find the slope of the line through $(3, -2)$ and $(-1, 6)$.

Problem 13

Write the equation of a line with slope $-\frac{2}{3}$ and y -intercept 4.

Problem 14

Convert to slope-intercept form: $3x - 4y = 24$

State m and b .

Problem 15

Find the equation of the line through the points $(1, 7)$ and $(4, -2)$.

Solutions

Part A: Foundations (Order of Operations & Properties)

Problem 1

Simplify: $8 + 12 \div 4 - 2 \times 3$

Solution:

Follow PEMDAS (division and multiplication before addition and subtraction):

$$\begin{aligned} &8 + 12 \div 4 - 2 \times 3 \\ &= 8 + 3 - 6 \\ &= 11 - 6 \\ &= 5 \end{aligned}$$

Answer: 5

Problem 2

Simplify: $(15 - 3)^2 \div 6 + 4$

Solution:

Follow PEMDAS (parentheses first, then exponents, then division, then addition):

$$\begin{aligned} &(15 - 3)^2 \div 6 + 4 \\ &= (12)^2 \div 6 + 4 \\ &= 144 \div 6 + 4 \\ &= 24 + 4 \\ &= 28 \end{aligned}$$

Answer: 28

Problem 3

Simplify by combining like terms: $8x - 3y + 2x + 7y - 5$

Solution:

Combine like terms (group x terms, y terms, and constants):

$$\begin{aligned} &8x - 3y + 2x + 7y - 5 \\ &= (8x + 2x) + (-3y + 7y) - 5 \\ &= 10x + 4y - 5 \end{aligned}$$

Answer: $10x + 4y - 5$

Problem 4

Simplify: $4(2x - 3) - 3(x + 5) + 7$

Solution:

Use the distributive property, then combine like terms:

$$\begin{aligned} &4(2x - 3) - 3(x + 5) + 7 \\ &= 8x - 12 - 3x - 15 + 7 \\ &= (8x - 3x) + (-12 - 15 + 7) \\ &= 5x - 20 \end{aligned}$$

Answer: $5x - 20$

Problem 5

A rectangular garden has length $(3x + 5)$ feet and width $(2x - 1)$ feet.
Write a simplified expression for the perimeter.

Solution:

Perimeter = $2(\text{length}) + 2(\text{width})$

$$\begin{aligned} P &= 2(3x + 5) + 2(2x - 1) \\ &= 6x + 10 + 4x - 2 \\ &= 10x + 8 \end{aligned}$$

Answer: $(10x + 8)$ feet

Part B: Equations

Problem 6

Solve: $5x - 8 = 2x + 13$

Solution:

$$5x - 8 = 2x + 13$$

$$5x - 2x = 13 + 8 \quad (\text{subtract } 2x \text{ from both sides, add 8 to both sides})$$

$$3x = 21$$

$$x = 7 \quad (\text{divide both sides by 3})$$

Answer: $x = 7$

Problem 7

Solve: $\frac{1}{3}(x - 6) = 0.5x + 4$

Solution:

Multiply both sides by 6 to clear fractions (LCD of 3 and 2):

$$6 \cdot \frac{1}{3}(x - 6) = 6(0.5x + 4)$$

$$2(x - 6) = 3x + 24$$

$$2x - 12 = 3x + 24$$

$$-12 - 24 = 3x - 2x$$

$$-36 = x$$

Answer: $x = -36$

Problem 8

Solve for h : $A = \frac{1}{2}bh$

Solution:

Multiply both sides by 2:

$$2A = bh$$

Divide both sides by b :

$$\frac{2A}{b} = h$$

Answer: $h = \frac{2A}{b}$

Part C: Inequalities

Problem 9

Solve: $-4x + 7 \leq -9$

Solution:

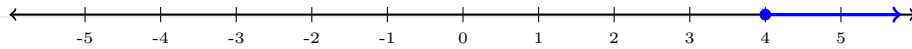
$$-4x + 7 \leq -9$$

$$-4x \leq -16 \quad (\text{subtract 7 from both sides})$$

$$x \geq 4 \quad (\text{divide by } -4; \text{reverse inequality sign})$$

Answer: $x \geq 4$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Answer: $[4, \infty)$

Problem 10

Solve: $-5 < 2x + 1 \leq 7$

Solution:

Solve the compound inequality by working on all parts:

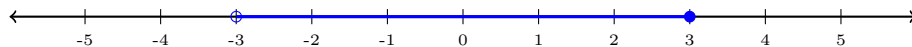
$$-5 < 2x + 1 \leq 7$$

$$-6 < 2x \leq 6 \quad (\text{subtract 1 from all parts})$$

$$-3 < x \leq 3 \quad (\text{divide all parts by 2})$$

Answer: $-3 < x \leq 3$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Answer: $(-3, 3]$

Problem 11

A streaming service costs \$8.99 per month. You have a \$50 gift card.

Write and solve an inequality to find the maximum number of whole months you can subscribe.

Solution:

Let m = number of months

The total cost must be less than or equal to \$50:

$$8.99m \leq 50$$

$$m \leq \frac{50}{8.99}$$

$$m \leq 5.56\dots$$

Since we need whole months: $m \leq 5$

Answer: Maximum of 5 months

Part D: Linear Functions & Lines

Problem 12

Find the slope of the line through $(3, -2)$ and $(-1, 6)$.

Solution:

Use the slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m = \frac{6 - (-2)}{-1 - 3}$$

$$m = \frac{8}{-4}$$

$$m = -2$$

Answer: $m = -2$

Problem 13

Write the equation of a line with slope $-\frac{2}{3}$ and y -intercept 4.

Solution:

Use slope-intercept form: $y = mx + b$

Given: $m = -\frac{2}{3}$ and $b = 4$

Substitute: $y = -\frac{2}{3}x + 4$

Answer: $y = -\frac{2}{3}x + 4$

Problem 14

Convert to slope-intercept form: $3x - 4y = 24$

State m and b .

Solution:

Solve for y :

$$3x - 4y = 24$$

$$-4y = -3x + 24 \quad (\text{subtract } 3x \text{ from both sides})$$

$$y = \frac{3}{4}x - 6 \quad (\text{divide by } -4)$$

Answer: $y = \frac{3}{4}x - 6$; $m = \frac{3}{4}$, $b = -6$

Problem 15

Find the equation of the line through the points $(1, 7)$ and $(4, -2)$.

Solution:

First find the slope:

$$m = \frac{-2 - 7}{4 - 1} = \frac{-9}{3} = -3$$

Use point-slope form with point $(1, 7)$:

$$y - 7 = -3(x - 1)$$

$$y - 7 = -3x + 3$$

$$y = -3x + 10$$

Answer: $y = -3x + 10$
